

Yikai WU

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EDUCATION

The Hong Kong Polytechnic University, Hong Kong SAR 09/2025 – 01/2027
(Expected)

M.Sc. in Urban Informatics and Smart Cities | **GPA: 3.89/4.30**

- **Core coursework:** Remote Sensing Image Processing, Urban and Geospatial Big Data Analysis, Advanced GIS, Advanced Mobile GIS and Location-Based Services

Southwest Jiaotong University, Chengdu, China 09/2021 – 06/2025

B.Eng. in Remote Sensing Science and Technology | **Average Mark: 84.44/100**

- **Core coursework:** Computer Vision and Pattern Recognition, Photogrammetry and Remote Sensing, 3D Laser Scanning, Digital Image Processing, Quantitative Remote Sensing

RESEARCH INTERESTS

Embodied AI for Autonomous Mapping; Structural Health Monitoring; 3D Vision; Geospatial AI

PUBLICATIONS

1. FAN W., XIAO J., **WU Y.**, et al. “A Review of Optimization Methods for Low-Altitude Aircraft Operations.” *Journal of Transportation Engineering and Information*, 23(3), 37–59, 2025. DOI: 10.19961/j.cnki.1672-4747.2025.06.044. (in Chinese)
2. **WU Y.** “Research Progress of Urban Ecological Evaluation Methods Based on Remote Sensing Indexes.” *Highlights in Science, Engineering and Technology*, 108, 27–33, 2024. DOI: 10.54097/rm8c6t44.
3. **WU Y.**, et al. “Target-level Multi-view RGB-D Quantification of Concrete Cracks and Spalling.” *In preparation*.
4. “From Photographic Diagnosis to Disease Trajectories: An Integrated Artificial Intelligence Framework for Oral Potentially Malignant Disorders.” *In preparation; anticipated second student author*.
5. ZHANG Y., **WU Y.**, AI R., ZHANG Q., LIN J., LIU Z., KANG L., AI R. “Machine Learning in Alzheimer’s Disease: A Comprehensive Review of Methods, Data Modalities, and Future Directions.” *In preparation*.

RESEARCH EXPERIENCE

Multi-view Perception and Quantification for Infrastructure Inspection 09/2025 – Present
Research Assistant | Joint Laboratory for Marine Infrastructure, The Hong Kong Polytechnic University
Supervisor: Dr. Rongpan Hu, Research Assistant Professor

- Investigated how to turn image-based defect detections into consistent physical measurements of concrete cracks and spalling, accounting for changes in viewpoint and visibility across observations.
- Developed a target-level RGB-D pipeline combining semantic segmentation, marker-based pose estimation, cross-view target association, and quality-gated multi-view fusion to measure defects in a shared metric reference frame.
- On **6 crack targets**, achieved MAE of **2.17 mm** for length and **0.22 mm** for mean width; on **2 spalling targets**, MARE were **2.90%** for area and **3.21%** for mean depth.

Identification of Advanced Support Conduits in Railway Construction

04/2023 – 05/2024

Scenarios

Project Leader | Student Research Training Program, VGE Laboratory

Advisor: Prof. CHEN Min, Southwest Jiaotong University

- Addressed the difficulty of checking conduit presence, position, and spacing during tunnel advance-support construction, where manual inspections provide limited quantitative records.
- Led dataset construction using **6,000+ field images** and trained YOLO-based detectors; integrated stereo-camera calibration and SGBM depth estimation to link image detections with 3D conduit locations.
- Achieved preliminary **AP@0.5 $\approx$ 0.80** and **AP $\approx$ 0.45** on the advance-support dataset; developed a detection–depth localization prototype and explored model pruning and ONNX deployment.

Urban Beauty and Distinctiveness Mapping from Street-view Imagery

2026 – Present

M.Sc. Research Project | GIFT Lab

Advisor: Prof. Xiao Li, The Hong Kong Polytechnic University

- Investigated how to identify urban scenes that are both widely perceived as beautiful and visually distinctive, rather than relying on average aesthetic scores that can obscure disagreement.
- Implemented a **seven-encoder voting ensemble**, 5,000-replicate bootstrap analysis, and multi-layer kNN distinctiveness scoring; connected image-level predictions to spatial aggregation and GIS exports.
- Completed inference on **56,530 Hong Kong panoramas (226,120 directional views)**, identifying **435 view-level beauty–distinctiveness candidates** for subsequent spatial analysis.

Historic building Reconstruction with NeRF

04/2024 – 07/2024

Student Research Training Program | Advisor: Prof. LIU Guoxiang, Southwest Jiaotong University

- Explored the use of NeRF for digitizing historic buildings from multi-view photographs.
- Applied NeRF to UAV and ground imagery collected at **Wenshu Monastery, Chengdu**.

High-resolution Gross Primary Productivity Estimation

04/2024 – 07/2025

Undergraduate Thesis | Advisor: Prof. WU Xiaodan, Southwest Jiaotong University

- Aimed to estimate gross primary productivity (GPP) at finer spatial resolution for local ecological analysis and future comparisons with coarser satellite products such as MODIS.
- Integrated HLS v2.0 imagery and FLUXNET observations in GEE, Python, and MATLAB; built a library of vegetation-specific SVM models and matched pixels to the corresponding models for estimation.
- Generated a fine resolution GPP map for the Chengdu region, providing a regional mapping output and a foundation for subsequent satellite-product comparison and ecological studies.

Machine Learning for Oral Disease Analysis and Medical Research

11/2024 – Present

Lead Researcher for Computational Implementation | Advisors: Prof. ZHANG Yimeng; Dr. AI Ruixue

- Addressed variation across oral-image datasets and the separation between image-based assessment and longitudinal follow-up, aiming to connect model evaluation with an integrated research platform.
- Validation-selected models achieved test Macro-F1 of **0.921** internally ($n = 665$), **0.885** on Dataset of OMD ($n = 173$), and **0.937** on OralTriD ($n = 80$); developed the web platform to bring oral-image assessment and follow-up information into one workflow.

AWARDS

“Challenge Cup” Third Prize ; Third Prize Scholarship — Southwest Jiaotong University